THERE ARE NO MAPS BETWEEN BOUNDED ABOVE COMPLEXES OVER A REGULAR RING WHEN THEIR SUPPORT RANGES ARE SEPARATED BY THE DIMENSION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 20766-20772 (canonical tag 0FXJ) and the counted proof is the proof environment at source lines 20774-20797, which immediately follows it and is its own proof.

Editorial note. The printed text uses the letter k both for the vanishing bound of the first complex and inside the phrase there exists an k; the wording is copied exactly as printed.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

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Allowed standard prerequisites (editorial).

- A bounded above complex may be replaced by its truncation above the vanishing bound in low degrees without changing the Ext groups in question. used at proof lines 20774-20778. Puts the representing complex in the normal form in which the top degree argument applies.
- A short exact sequence of complexes with the shift of a term in the middle gives a distinguished triangle, so the vanishing statement can be checked on the two outer pieces. used at proof lines 20778-20783. Reduces the statement by induction on the number of nonzero terms to the case of a single module placed in one degree.
- Over a regular ring of finite dimension every module has a projective resolution of length at most the dimension, and a map of complexes from a bounded complex of projectives concentrated in a range to a complex vanishing above a lower bound is zero. used at proof lines 20783-20796. Provides the final vanishing, which is exactly the statement for the single module case.

Source reference. Original lemma 00OE: sources/stacks-chow/tags.txt, source lines 892–892. The reference gives finite global dimension of the regular ring, used at proof lines 20784-20785 to bound the length of the projective resolution; the induction and the final comparison are inside the counted span.

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Source reference. Original lemma 064B: sources/stacks-chow/tags.txt, source lines 7860–7860. The reference identifies maps in the derived category with maps of complexes from a bounded complex of projectives, used at proof lines 20796-20797 to conclude the vanishing from the disjointness of supports; it is the last step and the preparation for it is inside the counted span.

Lemma 0.2. *Let R be a Noetherian regular ring of dimension $d < \infty$. Let $K, L \in D^-(R)$. Assume there exists an k such that $H^i(K) = 0$ for $i \leq k$ and $H^i(L) = 0$ for $i \geq k - d + 1$. Then $\mathrm{Hom}_{D(R)}(K, L) = 0$.*

Proof. Let $K^\bullet$ be a bounded above complex representing K, say $K^i = 0$ for $i \geq n + 1$. After replacing $K^\bullet$ by $\tau_{\geq k+1} K^\bullet$ we may assume $K^i = 0$ for $i \leq k$. Then we may use the distinguished triangle

$$K^n[-n] \to K^\bullet \to \sigma_{\leq n-1} K^\bullet$$

to see it suffices to prove the lemma for $K^n[-n]$ and $\sigma_{\leq n-1} K^\bullet$. By induction on n, we conclude that it suffices to prove the lemma in case K is represented by the complex $M[-m]$ for some R-module M and some $m \geq k + 1$. Since R has global dimension d by Algebra, Lemma 00OE we see that M has a projective resolution $0 \to P_d \to \ldots \to P_0 \to M \to 0$. Then the complex $P^\bullet$ having P_i in degree $m - i$ is a bounded complex of projectives representing $M[-m]$. On the other hand, we can choose a complex $L^\bullet$ representing L with $L^i = 0$ for $i \geq k - d + 1$. Hence any map of complexes $P^\bullet \to L^\bullet$ is zero. This implies the lemma by Derived Categories, Lemma 064B. □